

Use conversion rates to find each value. Round to the nearest thousandth.

U.S. Customary System to Metric System		
1 fluid ounce (fl oz)	=	28.413 milliliters (mL)
1 cup (c)	=	236.588 milliliters (mL)
1 fluid ounce (fl oz)	=	0.030 liters (L)
1 quart (qt)	=	0.946 liters (L)
1 gallon (gal)	=	3.785 liters (L)

Metric System to U.S. Customary System		
1 milliliter (mL)	=	0.034 fluid ounces (fl oz)
1 milliliter (mL)	=	0.004 cups (c)
1 liter (L)	=	33.814 fluid ounces (fl oz)
1 liter (L)	=	1.057 quarts (qt)
1 liter (L)	=	0.264 gallons (gal)

1. 4 cups = 946.352 milliliters 2. 350 milliliters = 12.318 fluid ounces

$$4(236.588)$$

$$\frac{350}{28.413}$$

3. 2 liters = 2.114 quarts 4. 8 gallons = 30.303 liters

$$2(1.057)$$

$$\frac{8}{0.264}$$

5. Mateo buys a package with 24 bottles of water. Each bottle holds 8 ounces of water. How many milliliters are in the package?

$$\frac{192}{0.034} = 5,647.059 \text{ mL}$$

6. Kristen is mixing sweet tea and lemonade to make her favorite beverage. She mixes 20 fluid ounces of sweet tea with 16 fluid ounces of lemonade. How many liters of the beverage did she mix?

$$\frac{36}{33.814} = 1.065 \text{ L}$$

7. How many milliliters is 5 quarts of water? Round your answer to the nearest tenth.

$$\frac{1,000 \text{ mL}}{1 \text{ L}} \cdot \frac{1 \text{ L}}{1.057 \text{ qt}} \cdot 5 \text{ quarts} = 4,730.4 \text{ mL}$$

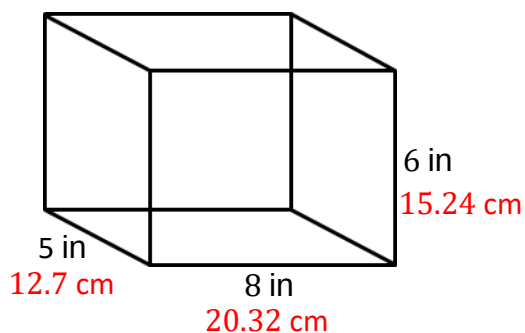
8. 7 L + 150 fl oz = 386.698 fl oz

$$7\text{L} = 236.698 \text{ fl oz}$$

9. 7 L + 150 fl oz = 11.436 L

$$150 \text{ fl oz} = 4.436 \text{ L}$$

10. If 1 inch = 2.54 centimeters, what is the volume of the cube below in cubic centimeters? Round your answer to the nearest tenth.



$$12.7(20.32)(15.24) = 3,932.9 \text{ cm}^3$$